Introduction :

The objective of this chapter is to study potential, denoted by V. The source of potential may be point charge or other types of charge configuration. Initially we shall obtain expression for the potential due to point charge and then this idea can be extended to obtain V due to other types of charge configurations.

Since the basic definition of scalar potential is

$$\text{Potential} = \frac{\text{Work done}}{\text{Charge}}$$

It is necessary to have knowledge of work done. In the next section we shall study work done and then go for the potential.

4.1 Work Done :

Energy expended is nothing but work done. In this article, we are going to calculate how much work is required to be done in moving a point charge in electric field.

We have the basic formula for work done

Work done = Force applied × displacement

A charge Q experiences a force in an electric field $\bar{E}$. If we attempt to move the charge against the electric field, we have to apply a force equal and opposite to that exerted by the field. This requires us to do work.

If we wish to move the charge in the direction of the field, work done is negative because we do not do the work, the field does.

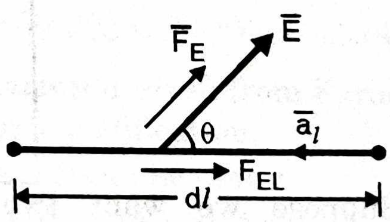

Fig. 4.1.1 : Finding work done

Suppose we wish to move a charge Q, a distance dl in an electric field $\bar{E}$ as shown in Fig. 4.1.1. The force on Q due to electric field is,

$$\bar{F}_E = Q\bar{E}$$

Subscript E indicates that this force is applied by electric field E, not due to charge.

The component of this force in the direction $d\bar{l}$ is :

$$F_{El} = F_E \cos\theta = \bar{F}_E \cdot \bar{a}_l = Q\bar{E} \cdot \bar{a}_l$$

When we are moving a charge Q a distance 'dl', then the force which we must apply is equal and opposite to the force due to field.

$$F_{appl} = -F_{El} = -Q\bar{E} \cdot \bar{a}_l$$

The work done is force multiplied by distance. The work done in moving a charge Q by differential length dl is a differential work, given by,

$$dW = F_{appl}\, dl = (-Q\bar{E} \cdot \bar{a}_l)\, dl$$

but $\qquad dl\,\bar{a}_l = d\bar{l}$

so that $\qquad dW = -Q\,\bar{E} \cdot d\bar{l}$

This is the **incremental works** done in moving a charge Q in the presence of the electric field $\bar{E}$ by a differential length dl.

The total work required to move the charge a finite distance between two points from initial position to final position is obtained by integrating dW, i.e.

$$W = \int dW$$

so that

$$\boxed{W = -Q \int_{\text{initial}}^{\text{final}} \bar{E} \cdot d\bar{l} \text{ (Joules)}} \qquad \ldots(4.1.1)$$

Work done is measured in Joules.

The expressions for $d\bar{l}$ in three coordinate systems are :

$$\boxed{\begin{aligned} d\bar{l} &= dx\,\bar{a}_x + dy\,\bar{a}_y + dz\,\bar{a}_z && \text{(Cartesian)} \\ d\bar{l} &= dr\,\bar{a}_r + r\,d\phi\,\bar{a}_\phi + dz\,\bar{a}_z && \text{(Cylindrical)} \\ d\bar{l} &= dr\,\bar{a}_r + r\,d\theta\,\bar{a}_\theta + r\sin\theta\,d\phi\,\bar{a}_\phi && \text{(Spherical)} \end{aligned}} \qquad \ldots(4.1.2)$$

Suppose we want two move a point charge Q from $x = x_2$ to $x = x_1$ in an electric field $\bar{E}$. There are different paths possible as shown in Fig. 4.1.2(a). Because the initial and final positions for these paths are same, work done must be same for any path choosen. That is, **work done is independent of the path.**

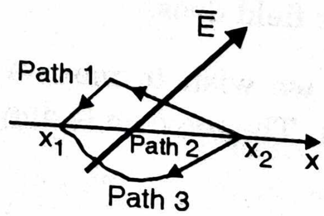

Fig. 4.1.2(a) : Work done along different paths

4.1.1 Work Done in a Loop (Conservative Property of an Electric Field) :

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Now suppose the charge is moved around any closed loop as shown in Fig. 4.1.2(b).

Because initial position is x_3 which is equal to final position, the work done is zero.

$$W = -Q \int_{initial}^{final} \bar{E} \cdot d\bar{l} = 0$$

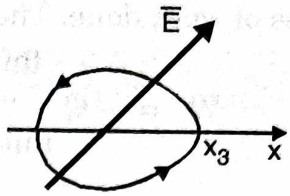

Fig. 4.1.2(b) : Conservative field

That is **work done in moving the charge around any closed loop is zero**. This can be written as,

$$\boxed{\oint \bar{E} \cdot d\bar{l} = 0} \quad \text{...(4.1.3)}$$

A small circle is placed on the integral sign indicates the closed nature of the path. The above equation is true for static fields. Such a field is called as **conservative** field or irrotational field. If an electric field varies with time, it need not be conservative.

Equation (4.1.3) is called as integral form of **Maxwell's equation** derived from Faraday's law for static field. For time varying field this expression requires modification. Faraday's law and its modification is studied in chapter on Maxwell's equations. (Chapter 8)

Applying Stoke's theorem to integral form in Equation (4.1.3)

$$\oint \bar{E} \cdot d\bar{l} = \int_s (\nabla \times \bar{E}) \cdot d\bar{s}$$

For conservative field when LHS of above equation is zero, then the RHS is also zero, giving

$$\int_s (\nabla \times \bar{E})\, d\bar{s} = 0$$

or

$$\boxed{\nabla \times \bar{E} = 0} \quad \text{...(4.1.4)}$$

This equations is referred to as point form of **Maxwell's equation** derived from Faraday's law for static field. For time varying fields this relation requires modification.

Note : There are two views that expression is Equation (4.1.3) can be seen :

(i) Since the work done in a loop is zero, the integral in Equation (4.1.3) is zero.

(ii) We know E has a unit of V/m and dl is expressed in meters. Combinely $\bar{E} \cdot d\bar{l}$ is having unit of (V/m × m) volts.

The integration is nothing but summation. Thus the integral of $\bar{E} \cdot d\bar{l}$ gives summation of voltages. The circle on the integral sign indicates summation in a loop, means $\oint \bar{E} \cdot d\bar{l} = 0$. indicates summation of voltages in a loop is zero. What is this statement for ? This is nothing but Kirchoff's Voltage Law (KVL).

4.1.2 Important Properties of Work Done :

Using the relations derived in the pervious section we can state important properties of work done. These relations are :

$$W = -Q \int_{\text{initial}}^{\text{final}} \bar{E} \cdot d\bar{l} \quad \text{...(i)}$$

$$W = -Q \oint \bar{E} \cdot d\bar{l} = 0 \quad \text{...(ii)}$$

Properties :

i) When the work is required to calculate while moving the point charge from point B to A then initial and final positions of charge are B and A respectively and Equation (i) becomes

$$W \text{ (from B to A)} = -Q \int_{B}^{A} \bar{E} \cdot d\bar{l}$$

ii) When the direction of path ($d\bar{l}$) is against the electric field then

$\bar{E} \cdot d\bar{l} = E\, dl \cos(180) = -E\, dl$. This negative sign makes the work done to be positive. Then the external source is said to be doing the work.

iii) But when the path selected ($d\bar{l}$) is in the direction of electric field then $\bar{E} \cdot d\bar{l} = E\, dl \cos(0) = +E\, dl$. This results in work done to be negative. Then it is said that the field itself is doing the work, no external source is required.

iv) The selected path, if perpendicular to field then $\bar{E} \cdot d\bar{l} = E\, dl \cos(90) = 0$. This makes work done to be zero.

v) From Equation (ii), when the charge is moved in a loop then the total work done is zero.

vi) The work done is independent of the path, it depends only on the initial and final positions of the path.

Important Formulae

$$W = -Q \int_{\text{initial}}^{\text{final}} \bar{E} \cdot d\bar{l} = 0$$

$$\oint \bar{E} \cdot d\bar{l} = 0 \qquad \nabla \times \bar{E} = 0$$